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Experiment No 8

Aim

To configure a cryptocurrency wallet using MetaMask and integrate it with a decentralized application (DApp) to perform and verify blockchain transactions.

Theory

MetaMask

MetaMask is a cryptocurrency wallet that provides users with an interface to interact with blockchain networks and decentralized applications. It manages blockchain accounts and allows users to approve or reject transactions initiated by DApps.

Decentralized Application (DApp)

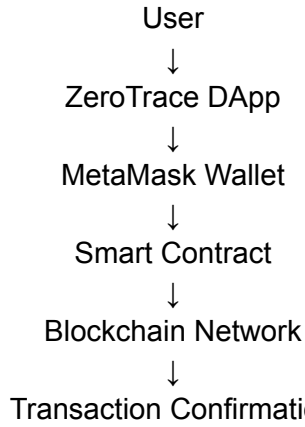
A Decentralized Application (DApp) is an application that interacts with blockchain-based smart contracts. The frontend of a DApp communicates with the blockchain through a Web3 provider such as MetaMask, which acts as a bridge between the user's browser and the blockchain network.

ZeroTrace DApp

In this experiment, the ZeroTrace DApp, a privacy-preserving quadratic funding platform, was used to demonstrate MetaMask integration. The DApp uses Web3 technologies to connect the user's wallet and interact with deployed smart contracts on the Sepolia test network. After connecting MetaMask, a blockchain transaction (submitting a project) was initiated through the DApp. MetaMask displayed the transaction request, which was confirmed by the user. The resulting transaction was then verified using its transaction ID on a blockchain explorer.

Interaction Flow

The basic interaction can be represented as:



MetaMask Functions and Transaction Hash

- **Account management:** creates and holds the accounts used to sign transactions.

- **Network selection:** connects to networks such as Sepolia.
- **Transaction signing:** the user reviews and approves each request before it is sent.
- **Connection authorization:** the user decides which website may view the account and request transactions.

A transaction hash (transaction ID) is generated for every submitted blockchain transaction and can be used to verify its status and details on a blockchain explorer such as Etherscan.

Outputs

Fig 1: ZeroTrace DApp home page (privacy-preserving quadratic funding platform)

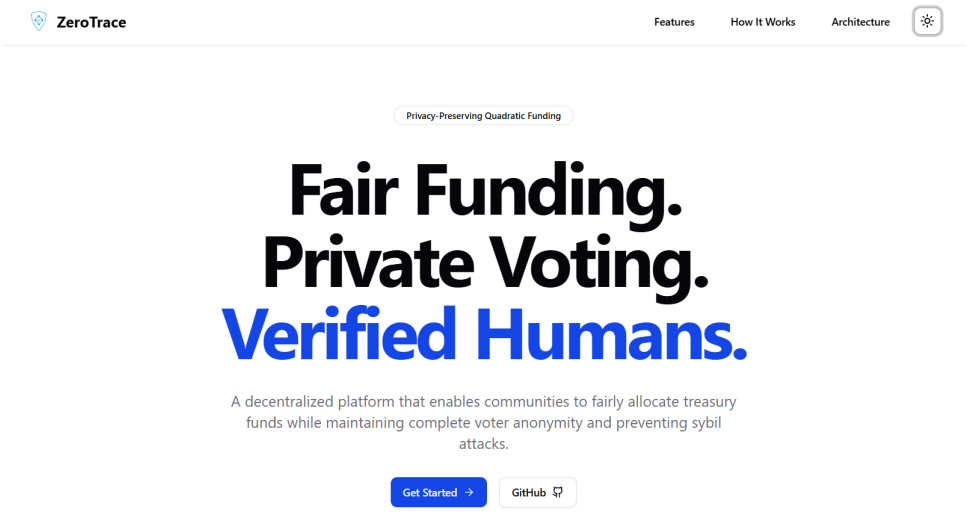


Fig 2: "Connect Your Wallet" page of the ZeroTrace DApp with the Connect Wallet option

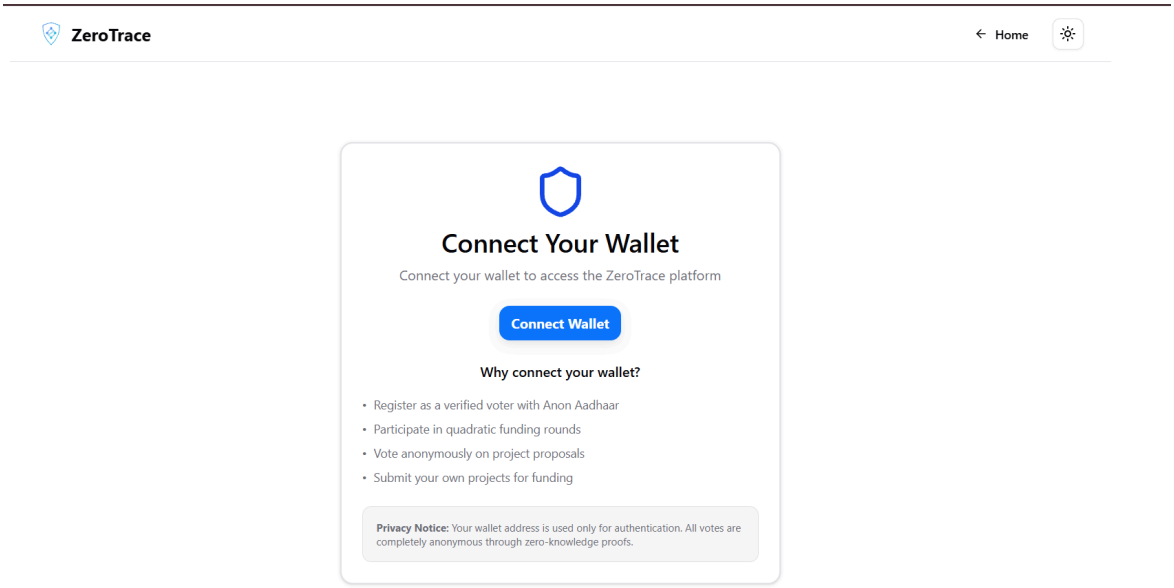


Fig 3: Selecting MetaMask in the "Connect a Wallet" dialog and approving the connection request in MetaMask

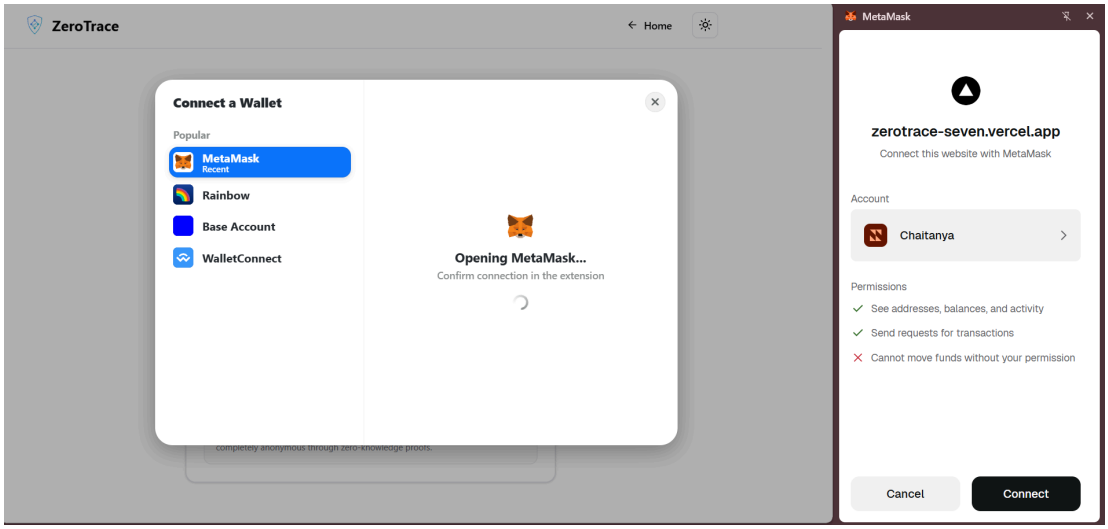


Fig 4: Submitting a new project in the DApp and the MetaMask transaction request (Sepolia network) awaiting confirmation

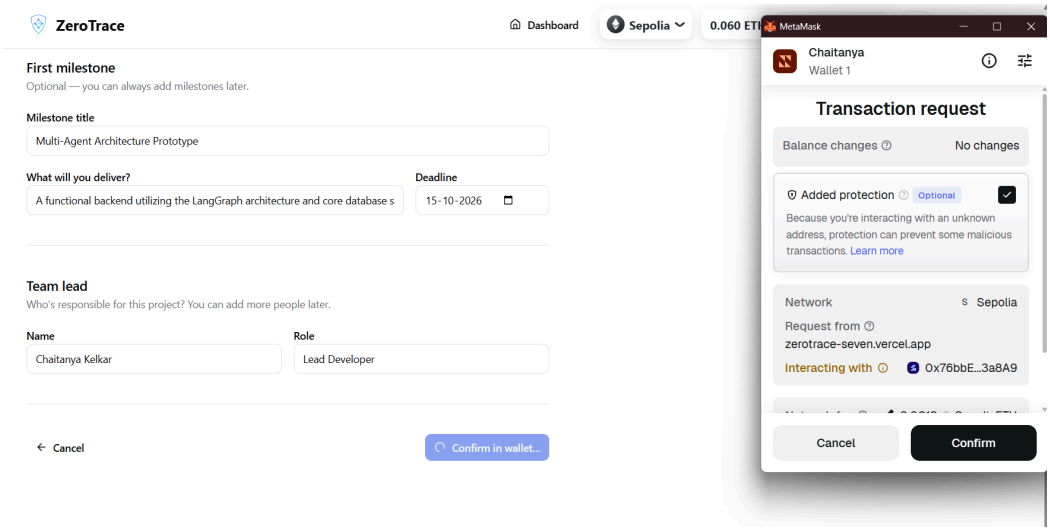
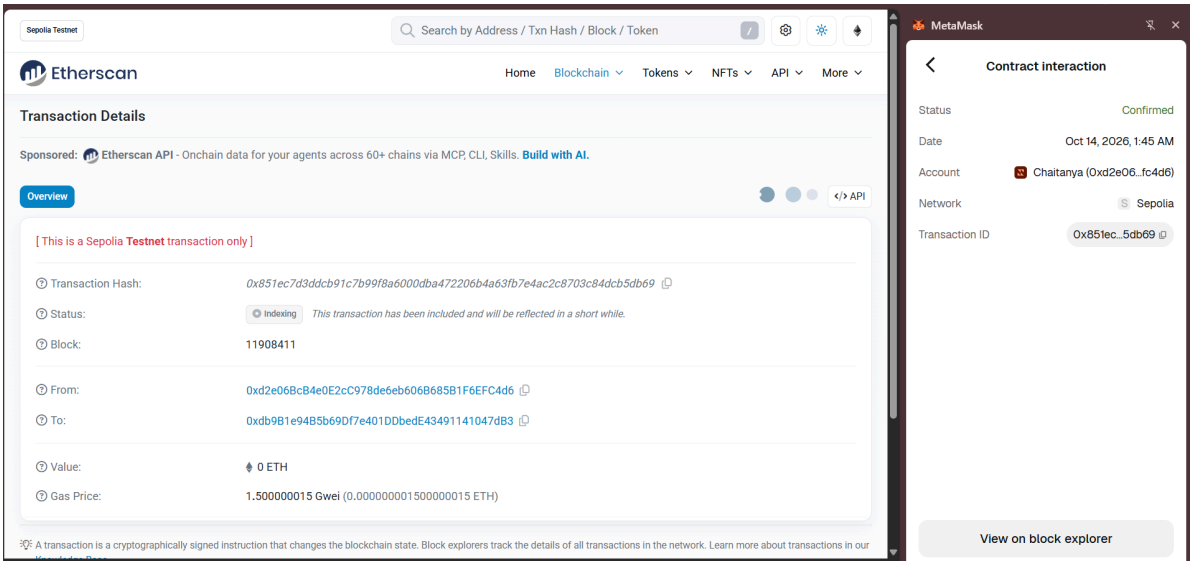


Fig 5: Transaction verification: MetaMask shows the contract interaction as Confirmed, and the transaction ID is verified on the Etherscan Sepolia block explorer



Conclusion

Thus, the MetaMask cryptocurrency wallet was successfully configured and integrated with the ZeroTrace DApp. The DApp was able to connect to the MetaMask wallet and initiate a blockchain transaction on the Sepolia test network, which was approved by the user through MetaMask. The transaction was successfully recorded and verified using its transaction ID on the Etherscan blockchain explorer. This experiment demonstrated the interaction between a cryptocurrency wallet, a decentralized application, smart contracts, and a blockchain network.